UNIT 1 SPECIAL TOPIC ANALYSES (SPM 5708)

In Unit 1, we learned basic R functions and data manipulation techniques along with a few statistical methods that are used in sports analytics. These methods include *t*-tests, Chi-Square tests, *z*-scores (standardization and normalization), and correlations.

Over the following pages, you will be asked to apply these techniques to a series of real-world datasets and communicate your findings. While you are welcome to seek your classmates out for assistance, you are not allowed to directly copy another person's answers. You must show all your own work in the Word document and code you submit.

Please note that I do not explicitly tell you which method to use in many of these problems. That is because I want you to think for yourself about *why* you are choosing a method to analyze a given dataset and *how* you can best approach the problem at hand. Keep in mind that I often ask for detailed interpretations of your findings from *both* a statistical *and* practical perspective.

Please **save all your codes in an annotated R Script** that you will submit, along with your answers in this Word document, to the *Special Topic Analyses 1* drop box on Canvas. Please use a different color font when typing your answers in this Word document.

1.1. Rookie Improvement

The assertion is frequently made that rookies need time to adjust to the professional ranks. In essence, it is believed that there is a learning curve that will see them struggle through some of their early games before reaching their full potential. But, *on average*, is this actually the case?

1. Using *at least* two football stats of your choice and the appropriate statistical test, test whether QBs' *average* performances are significantly higher in their second seasons than their first seasons. Report the findings of your results below, making sure that you communicate them properly. You will need to use the *Rookies* and *SecondSeason* sheets in the *SecondYear* dataset for this analysis. To find out more about what the various stat columns mean, check out this glossary of terms on pro-football-reference (8 pts).

   ANSWER: Note: For this exercise and the others, we will use a significance level (alpha) equal to 0.05.

   To be able to answer the question of whether players, in this instance, QBs, have better performances on average in their second season or not, we will use a one-tail independent Sample T-test (with direction). Our null hypothesis, H_0: $\mu_d <= 0$, and our alternative hypothesis is that H1: $\mu_d > 0$. We will use the AdjNYPA and Rate statistics to determine performance change.

   Starting with AdjNYPA, after performing the t-test, we get a p-value equal to 0.00631, so we reject the null hypothesis. And we can assume that players have a better adjusted net yard average in their second year than in the first year.

   For Rate, we get a p-value equal to 0.002954 so we also reject the null hypothesis, we can then assume that players are better in their second year than the first year.

2. Why did you choose to examine the stats that you did? Is there a way you could have assessed which statistics were most strongly related (relevant) to a team's performance? You will earn 3 bonus points if you show your work here (4 pts).

   ANSWER: I chose those stats based on their descriptions and their relevance for QB position. Rate because it is made to rate passing, and in our data set all the players are quaterbacks. Additionally, I looked at the correlation between variables and team's performance for the stats included (only a few of them relevant to the QB position in my opinion), I would have still chosen AdjNYPA and Rate which both had a correlation greater than 0.48 compared to winning percentage which I manually added to the data.

   I manually computed Winning percentage (WPct) to be able to compare the correlation between WPct and the variables in the data set by adding loses and wins for each player and divided wins by the total games played. I got the correlation coefficients below:

```
> cor(Rookie$WPct, Rookie$AdjNYPA)
[1] 0.5420927
> cor(Rookie$WPct, Rookie$Rate)
[1] 0.4807307
> cor(Rookie$WPct, Rookie$GS)
[1] 0.2983803
> cor(Rookie$WPct, Rookie$CmpPct)
[1] 0.381343
> cor(Rookie$WPct, Rookie$TDPct)
[1] 0.3973681
> cor(Rookie$WPct, Rookie$YPG)
[1] 0.06091271
> cor(Rookie$WPct, Rookie$AdjYPA) # it would
be the same as ADJNYPA but that one is net
[1] 0.5317966
```

```
> cor(Second$WPct, Second$AdjNYPA)
[1] 0.5965305
> cor(Second$WPct, Second$Rate)
[1] 0.6146037
> cor(Second$WPct, Second$GS)
[1] 0.4302563
> cor(Second$WPct, Second$CmpPct)
[1] 0.5364649
> cor(Second$WPct, Second$TDPct)
[1] 0.5295835
> cor(Second$WPct, Second$YPG)
[1] 0.2042433
```

1.2. Peyton Manning vs. Tom Brady

Peyton Manning and Tom Brady are two quarterbacks who will surely end up in Canton as first ballot Hall of Famers. And while there is no debating Tom Brady's seven Super Bowl rings in comparison to Peyton's two, there is still debate as to which of these modern era quarterbacks was the strongest from a statistical perspective during the seasons in which they crossed over during their primes.

The *BradyvsManning* Excel file contains passing stats for every QB who started at least eight games over the 2007 to 2014 NFL seasons. These seasons were chosen because they represent years during which Brady and Manning were both full-time starters in the league. You will be using this dataset to find how Brady and Manning's MVP seasons compared to their colleagues across the *ANYPA* (adjusted net yards per passing attempt), *CmpPct* (completion percent) and *AV* (approximate value) metrics.

1. During the timeframe of this dataset, Brady was MVP in 2007 and 2010, and Manning was MVP during the 2008, 2009, and 2013 seasons. For both QBs' MVP seasons (5 total), find the percentage of values (using all QBs from 2007-2014) that their respective stats in the *ANYPA*, *CmpPct*, and *AV* categories were higher than. Create a table displaying these results and paste it below (8 pts).

Player	Year	ANYPA_norm	CmpPct_norm	AV_norm
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
Tom Brady	2007	0.992	0.961	0.998
Tom Brady	2010	0.971	0.855	0.941
Peyton Manning	2008	0.781	0.902	0.874
Peyton Manning	2009	0.902	0.960	0.912
Peyton Manning	2013	0.992	0.948	0.962

2. Relative to their peers, which Brady or Manning MVP season appears to have been the most dominant during this timeframe? Justify your claim with statistical evidence (4 pts).

   ANSWER: The best season is Tom Brady's 2007 which is superior to his season in 2010 and superior to Manning's best season in 2013 by some decimal points in the variables CmpPct and AV.

1.3. Late Rounders vs. Early Rounders

Running backs in the modern NFL are often taken for granted. In 2018, Todd Gurley and Le'Veon Bell were effectively replaced by lesser-known backups in the games they missed, as were Kareem Hunt and Melvin Gordon. Some GMs and media members have been arguing that wasting an early-round pick on a RB like Ezekiel Elliott or Saquon Barkley is not worth it since equal production at this position can (supposedly) be found later in the draft or via free agency. Again, we can run statistical tests to gather evidence for or against this claim.

1. Looking at the *RunningBacks* dataset that contains career statistics (through 2018) for RBs selected in rounds 1-7 of the 2000-2018 NFL Drafts, conduct the appropriate analyses and state your claim for whether or not the average performances of first-round running backs are significantly *better* than the averages of RBs selected in other rounds. You can compare first-round RBs to RBs drafted in later, individual rounds, or groups of rounds, so long as you provide a rationale for the comparisons you make. The football stats you choose to

compare these RBs' averages across are also your choice, but you need to choose *at least three* and briefly explain why you chose them.

Note that it might be appropriate to calculate and create columns that have per-game or per-attempt stats (e.g., yards-per-rush) on your own. Further note that it might be helpful to filter your data to remove players who have not met certain thresholds you deem important. The column labeled *Rnd* shows the draft round a RB was selected in. Report/describe all your results *and* justify the groups and stats you compared below (8 pts).

ANSWER: We decided to compare it to the last two rounds available in the data set because we can drastically see whether there is a difference or not. We got rid of the players who had less than 17 games, because of that number being the amount of games played in the regular season. We also got rid of the players in the data set that did not have any rushes. We also got rid of the players with no receiving attempts. We ended up with 261 observations instead of the original 392. We will use an independent sample T-test because we will create two data sets, each including the players in the respective rounds. The stats, which had to be computed and added manually to the data set, used to compare players from the 1st round to the 6th and 7th round were:

- YPrA: Yards per Rush Attempt. I chose this statistic because it tells us how efficient the running back is, it allows us to compare all players, the higher the better.
    - Null Hypothesis: $\mu 1 = \mu 2$
    - Alternative Hypothesis: $\mu 1 \neq \mu 2$
    - P-value: 0.03101
    - Conclusion: We reject the null hypothesis. There seems to be a clear difference between 1st round and 6th and 7th round picked players for Rushing Yards per Attempt.
- RYPrA: Receiving Yards per Attempt. I believe that this will quantify the duality of a running back, paired with high Yards Per Rush Attempt, the player could be a double threat.
    - Null Hypothesis: $\mu 1 = \mu 2$
    - Alternative Hypothesis: $\mu 1 \neq \mu 2$
    - P-value: 0.1903
    - Conclusion: We fail to reject the null hypothesis. This could indicate that in the receiving area, later round picks' performance is like 1st round.
- TDR: This is the rushing touchdowns per Rush Attempt. I chose this stat to calculate how effective Running Back is at getting a positive outcome from possession.
    - Null Hypothesis: $\mu 1 = \mu 2$
    - Alternative Hypothesis: $\mu 1 \neq \mu 2$
    - P-value: 0.006941

- Conclusion: We reject the null hypothesis. There seems to be a clear difference between 1st round and 6th and 7th round picked players for rushing touchdowns per rush attempt.

2. What are some additional variables, beyond draft position, that may influence and/or predict a running back's production in the NFL? Elaborate on your suggestions (4 pts).

   ANSWER: Based on the information that I have gathered in this homework about this topic, I believe that turnovers and/or fumbles need to be considered. This could tell a story about ball security and increase or decrease the coach's confidence in the player's ability to protect the ball.

1.4. Ticketholder Status and Recall Ability

A member of the FSU athletics promotions team has conducted a survey to assess fans' abilities to recall which sponsoring corporations were associated with certain promotional events that took place during a recent football game. More specifically, the individual wanted to see whether recall of these advertisements was associated with the status of the ticket holder. The two statuses being looked at were *season ticket holders* and *single-game ticket holders*. The two promotions being investigated for recall were the *Steak & Shake* follow the number game and the *Farm Bureau* t-shirt toss. The *FSUPromotions* dataset displays the results of this hypothetical survey, with the 60 (30 of each status) surveyed attendees indicating which ticket category they fell into (*Status*) and whether they recalled (Yes/No) the specific promotion (*Recall*). Take note that the survey results for each promotion are recorded in separate Excel worksheets.

1. Which statistical method would be most appropriate for assessing whether a person's ability to recall a promotion is *independent* of their ticketholder status? Conduct this test for both promotions and interpret your results below (12 pts).

   ANSWER: We will conduct a chi-square test using a contingency table. Where our null hypothesis: The variables Status and Recall are independent, and our alternative hypothesis: The variables Status and Recall are associated.

   We first start with just the data for the SteakShake promotional event. We get a p-value equal to 0.0159 which leads us to reject the null hypothesis that the variables are independent. Meaning that status and ability to recall the SteakShake promotional event are related.

   Second, we have the data from the FarmBureau promotional event. We get a p-value equal to 0.2993 which is greater than our significant level 0.05 and we fail to reject the null hypothesis. We could assume that the Recall ability for FarmBureau promotional event is not related to the person's status.

   We also wanted to see if we were to put both data sets together what would happen given that normally, a bigger sample leads to more accurate outcomes. And when we put both data sets together using rbind, we got a p-value equal to 0.01032 which is again less than our significance level so we reject the null hypothesis that the variables are independent.

We could then assume that people's ability to recall both promotional events is related to their status.

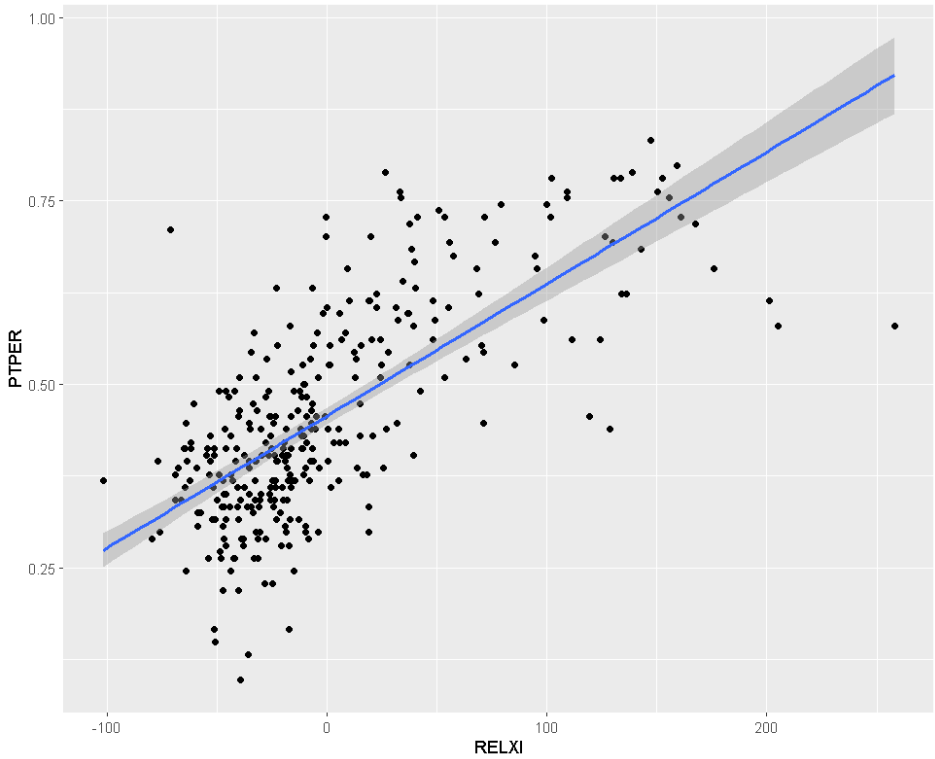
1.5. Money and Success in Both Types of Football

Do the wealthiest teams tend to be the most successful teams in both European football (soccer) and American football? That is, do clubs that spend more on playing talent compared to the rest of their opponents appear to end up performing better? The answer to such a question might be useful for team owners, general managers, and economists interested in this area.

1. Say you wanted to show that there is a linear relationship between how much money a team spends on players above or below the league average, and the winning percentage that team will have accumulated at the end of a season. What statistical test could help us assess this? Conduct this test on the *EPL* (English Premier League) worksheet that is housed in the *Payrolls* file and report your findings below (8 pts).

   Note that the column labeled *RELXI* contains the amount of money, in millions, that the team spent on transfer fees for the 11 players who appeared most frequently in their starting lineups, relative to the league average for that season. In addition, the *PTPER* column is the club's point percentage from that season, which is synonymous with win percentage (in EPL soccer, a win is worth 3 points and a draw 1, so it's calculated as the percentage of total available points that a team claimed rather than the number of wins divided by total games played). These should be your two columns of interest in this analysis.

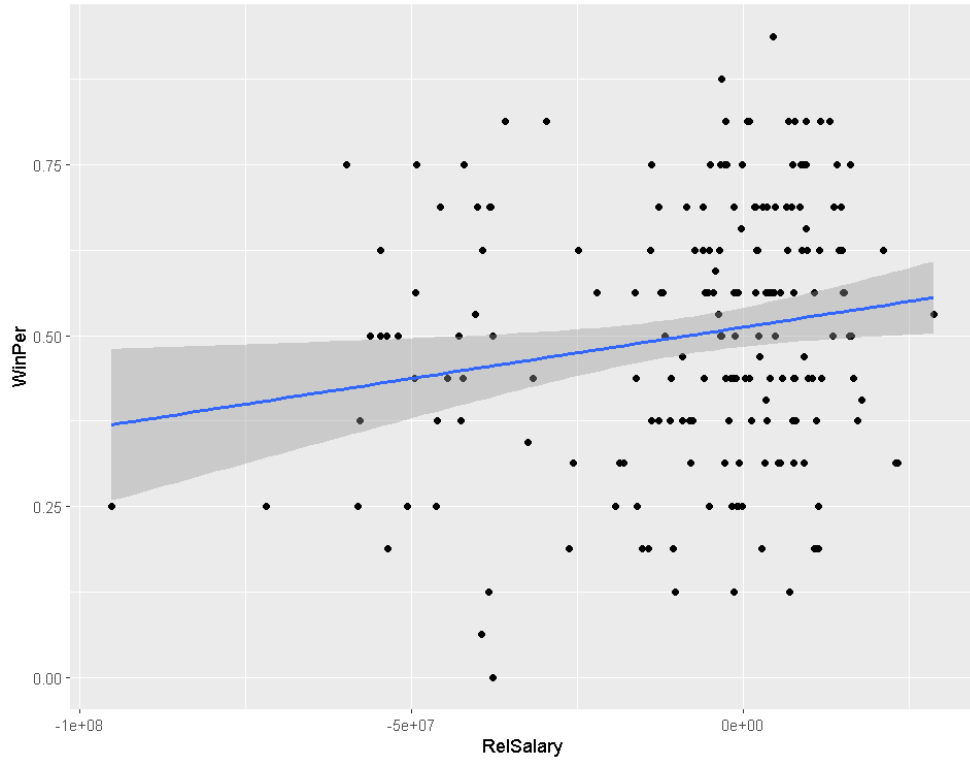
   ANSWER: We can use correlation to help us answer this question since this will show you if there is a linear relationship between our two variables. After performing correlation in R, we get a correlation coefficient equal to 0.7088591, which leads us to believe that there is a strong relationship between these two variables. Another way to assess the relationship between RELXI and PTPER would be to create a graph.



From our Scatterplot above we can see that while there is not a clear linear relationship, PTPER increases on average as RELXI increases, so with just doing a correlation test and graphically, we could assume that the more expensive a team is the more points a team will be able to make.

2. Is this relationship as strong in American football? Use the *NFL* worksheet in the *Payrolls* file to conduct the same analysis. Again, we are more interested in how much a team spent on player salaries relative to the league average for that season (*RelSalary*) and the relationship this has to win percentage (*WinPer*). Communicate your results and describe what they suggest (8 pts).

   ANSWER: When doing a correlation test for the NFL, we get a correlation coefficient equal to 0.171251 which leads us to assume that the relationship between high salaries and winning percentage is almost non-existent. Nonetheless, as in the graph below we see a fitted line that increases slightly from left to right as salary increases.



3. Compare the results of the two analyses you just ran. Speculate on why you obtained the results you did, remembering (or perhaps researching) the unique differences that exist between the salary rules and regulations in these two leagues (4 pts).

   ANSWER: The results obtained could perhaps be due to a salary cap in the NFL while in football (EPL) there is no limit to how much a team can pay its players. Nonetheless, in the EPL, this could change in the future, if not with a salary cap, with Squad Cost Ratio (SCR) and Sustainability and Systemic Resilience (SSR), explained here.

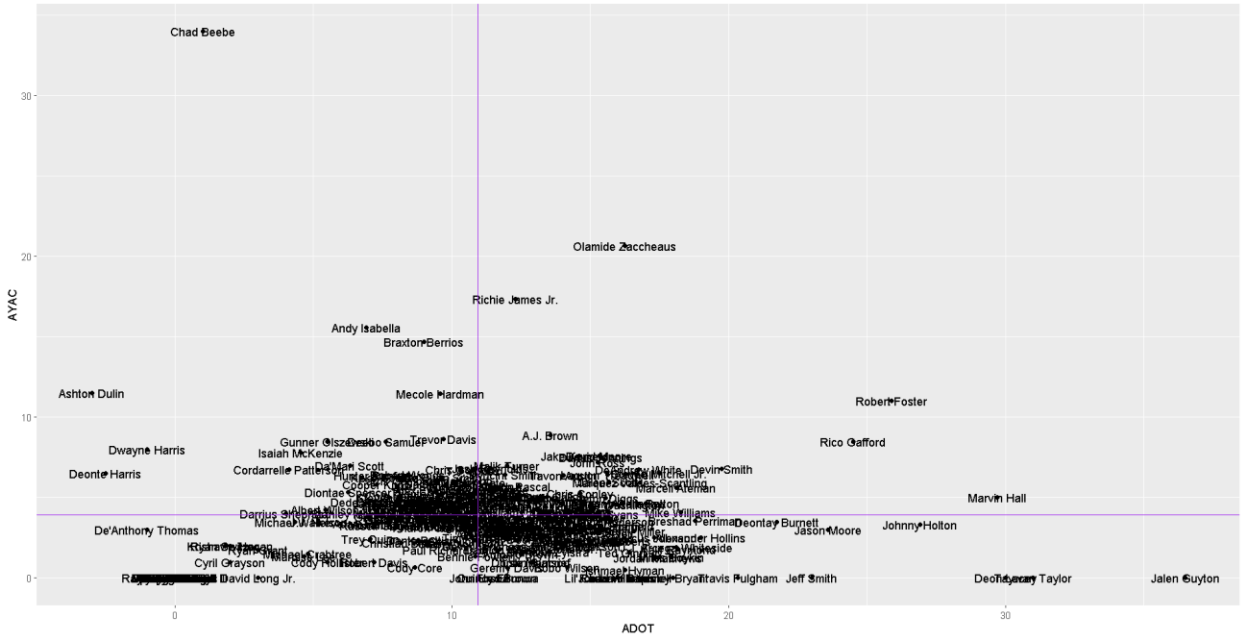
1.6. Using Scatterplots to Examine Wide Receiver Production

The *WR2019* dataset contains a list of all NFL wide receivers who played at least four games during the 2019 NFL regular season. Along with these receivers' names are two season-level stats—*ADOT* (*average depth of target*, the average number of yards from the line of scrimmage that passes targeting the receiver were thrown) and *AYAC* (*average yards* gained *after* a *catch*).

1. Find the correlation between *ADOT* and *AYAC* and interpret it below? Do you find this result surprising? Why or why not (8 pts)?

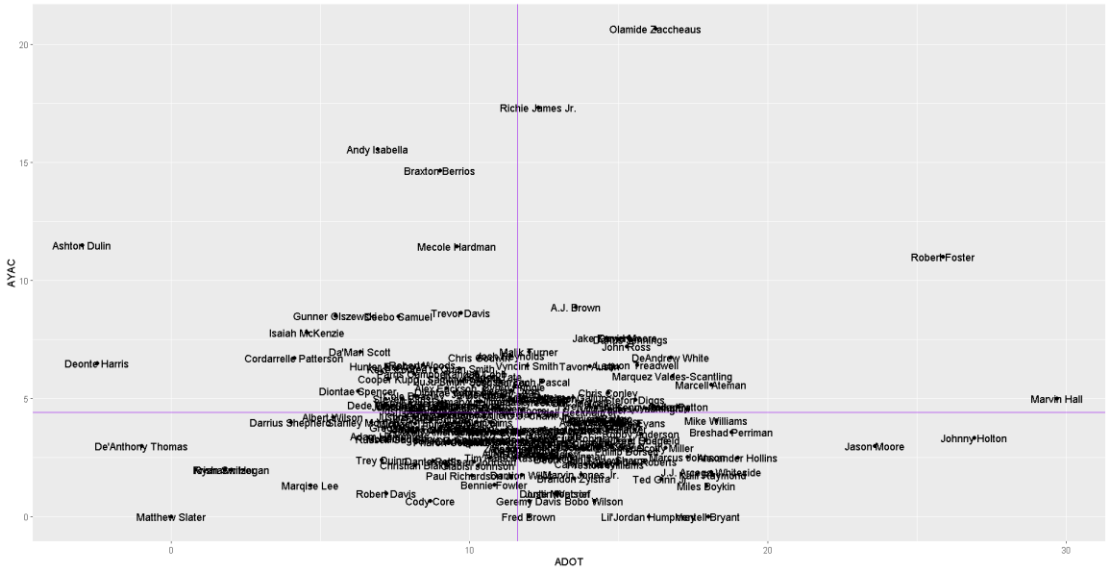
   ANSWER: The correlation coefficient is 0.03374329, this tells us that there is almost no relationship between these two statistics. No, I don't think it is surprising because they are both talking about different moments in the play as a wide receiver.

2. Create a scatterplot visualizing this relationship with *ADOT* on the x-axis and *AYAC* on the y-axis. Then, insert horizontal and vertical lines at the averages for *ADOT* and *AYAC,* and add labels for the players' names. Paste this final plot below (6 pts).

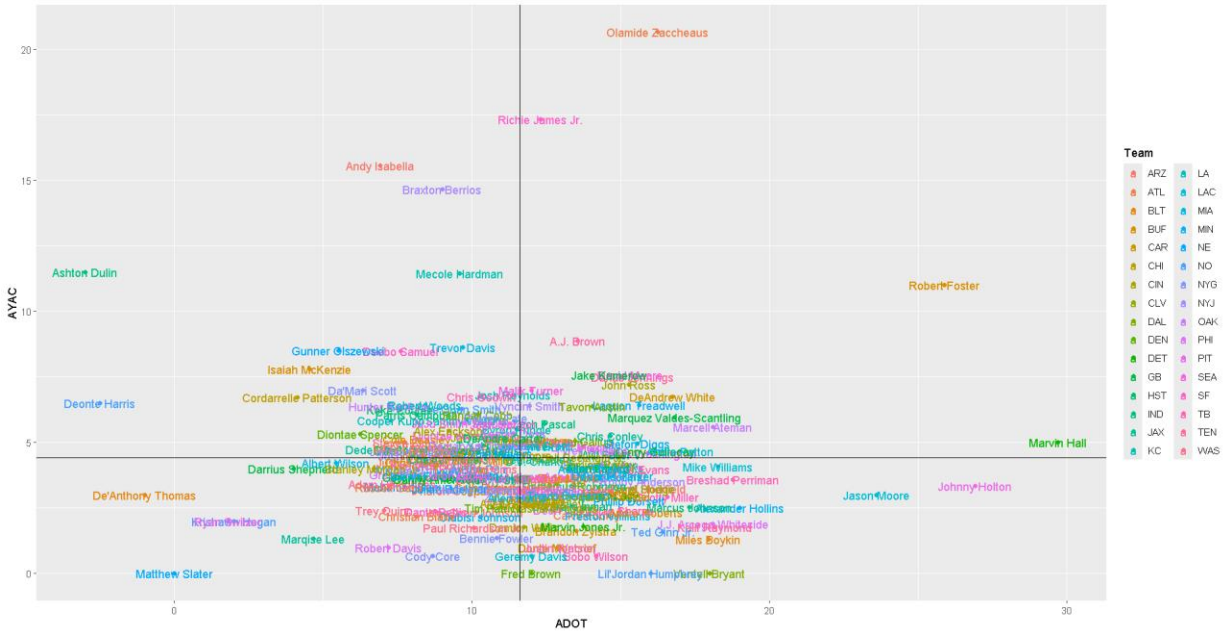


3. Could you use other variables in the data to set a specific threshold for inclusion or add another visual element to the chart? Create an enhanced plot and paste it below (6 pts).

ANSWER: We could use the variable Games to filter our players. We will filter the data so it only shows us Wide receivers that have played 5 games or more.



Or, we could also add color based on the team the player is from, just like this:



4. How could you use this chart to classify receivers in terms of where their yardage productivity comes from? In what other area of the sport industry might a chart like this be useful (4 pts)?

   ANSWER: This chart will facilitate the identification of the players that are above average whether in the ADOT (*average depth of target*) variable, or in the *AYAC* (*average yards* gained *after* a *catch*) variable, or in both variables.

   This chart could help coaches decide what kind of player based on these two statistics would be more appropriate to put on the field depending on the play or the situation. Like for example, if the coach know he is playing a good short defense but the players are not as fast as a specific wide receiver, he could create a play so that the wide receiver has the best chance of making the most yards after a catch.

   I think that a graph like this could be useful in any situation where you want to visualize either names, or teams, or products that are above or below average in two different categories. This could be useful, for example for the ticketing team to see what game days have the highest/lowest revenue and highest/lowest attendance. They could even make it a month-by-month situation where they assess how they can better adapt to attract the crowds, like for example in February around valentine's day a promotional event where you buy one and get the second discounted. It could also help a college basketball team to evaluate where their players stand compared to others in the same position at the conference the team is in.

1.7. Esports Affecting Actual Performance?

A lighthearted Reddit post by an NFL fan during the 2021-22 offseason highlighted what some have perceived to be a potential risk for professional athletes who enjoy video games as a hobby. The claim being made in the post was that Arizona Cardinals' quarterback Kyler Murray was performing statistically worse in NFL games that coincided with weekends

where *Call of Duty (COD)*, a popular video game that he played, was running promotions that incentivized long hours of play. But was this actually the case?

1. The *KylerCOD* dataset contains a series of traditional and advanced QB metrics from Murray's games during the 2019, 2020, and 2021 NFL seasons that coincided with weekends where special events were taking place on Call of Duty. We need to see if Murray performed significantly *worse* during these games compared to his known *career averages* in the categories of *CAYPA* (completed air yards—how far the ball traveled in the air—per completed pass attempt), *BadPct* (the percentage of pass attempts classified as poor/bad throws), and *YPScram* (yards per scramble). His career averages in these categories were 3.8 (*CAYPA*), .161 (*BadPct*), and 7.7 (*YPScram*), respectively. Higher values of *CAYPA* and *YPScram* are considered better, while lower values of *BadPct* are better. You also need to calculate his standard completion percentage *(Cmp/Att)* in these games to see if it was significantly lower than his career average of .667. After conducting the appropriate tests, did you find sufficient evidence that Murray was indeed performing worse than his career average in the sample of games that coincided with the Call of Duty events? How did you arrive at this conclusion? Answer these questions below (8 pts).

   ANSWER: His performance was not worse on the sample of games that coincided with the Call of Duty events as we will see below.
   We reached the statement above by doing one sample t-tests on each variable. For the variables CAYPA, BadPct, YPScram and the created variable CmpPct (standard completion percentage).

   CAYPA
   - Null hypothesis: CAYPA mean = 3.8
   - Alternative hypothesis: CAYPA mean != 3.8
   - p-value: 0.5971
   - conclusion: We fail to reject the null hypothesis that the mean of CAYPA is equal to 3.8

   BadPct
   - Null hypothesis: BadPct mean = .161
   - Alternative hypothesis: BadPct mean != .161
   - p-value: 0.4915
   - conclusion: We fail to reject the null hypothesis that the mean of BadPct is equal to .161

   YPScram
   - Null hypothesis: YPScram mean = 7.7
   - Alternative hypothesis: YPScram mean !=7.7
   - p-value: 0.5433
   - conclusion: We fail to reject the null hypothesis that the mean of YPScram is equal to 7.7

   CmpPct

- Null hypothesis: CmpPct mean = .667
- Alternative hypothesis: CmpPct mean != .667
- p-value: 0.8335
- conclusion: We fail to reject the null hypothesis that the mean of CmpPct is equal to .667

This completed assignment is due to the drop box on Canvas by Thursday, Feb. 19, before class starts. Remember to also upload a separate file of the R Script you used to solve each of these problems. Please annotate each segment of code to help identify the problem set it was run for.